We will prove it by induction that

$$E(T(n)) \leq c\, n \log n.$$

Choose c large enough such that the result holds for a base case.

Assume the result holds for all numbers $< n$

(Stronger induction $n=1$ $< n$ $= n$.)

To prove the result for n.

$$E(T(n)) = \sum_{k=2}^{n-1} \frac{2}{n} k \log k + \Theta(n)$$

Substituting in (1)

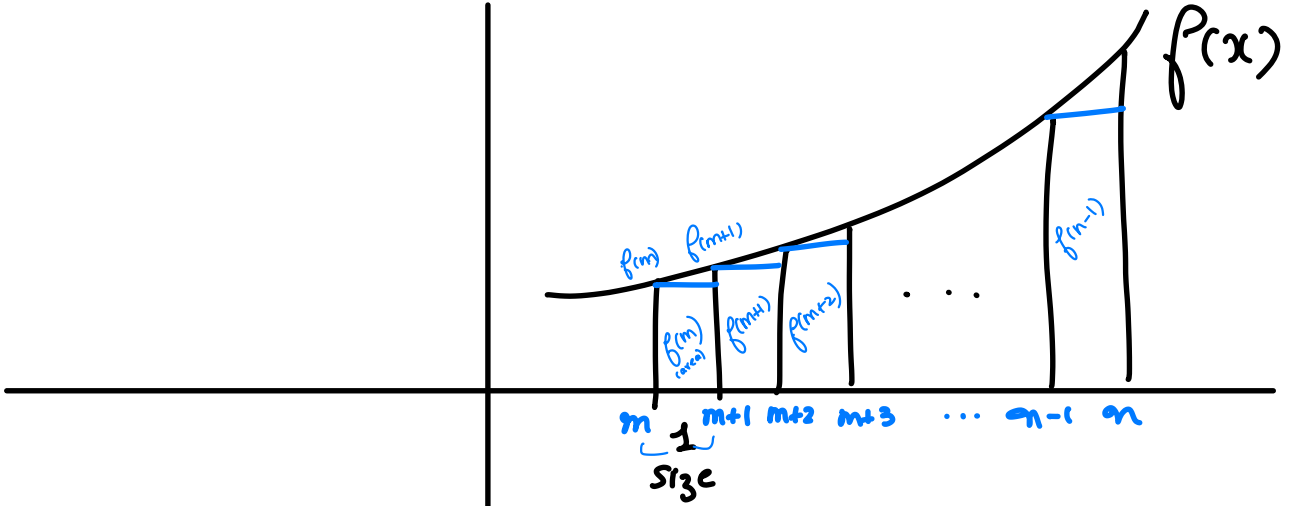
(by shifting $k=0, k=1$ cases inside $\Theta(n)$)

$$= \frac{2}{n} \sum_{k=2}^{n-1} k \log k + \Theta(n)$$

discrete → convert to continous by integration

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Let $f(x) = x \log x$ is an increasing function as $x \to \infty$



Sum of the rectangles $\leq$ Area under the curve $f(x)$ from m to n

$$\sum_{k=m}^{n-1} f(k) \leq \int_{x=m}^{x=n} f(x)\,dx$$

$$E(T(n)) = \left[\frac{2C}{n} \sum_{k=2}^{n-1} k \log k \right] + \Theta(n) \quad -(*)$$

$f(x)$ → monotonically increasing.

$$\sum_{k=m}^{n-1} f(x) \leq \int_{m}^{n} f(x)\, dx \quad - (I)$$

Put $n=2$ in (I) and substituting in (*),

$$E(T(n)) = \left[\frac{2c}{n} \sum_{k=2}^{n-1} k \log k \right] + \theta(n)$$

$$\leq \frac{2c}{n} \int_{2}^{n} x \log x \, dx + \theta(n)$$

$\int I \cdot II\,dx = I\int II\,dx - \int \frac{d}{dx}I \int II\,dx$

$$= \frac{2c}{n}\left[\log x \int x\,dx - \int\left(\frac{1}{x}\int x\,dx\right)dx\right] + \theta(n)$$

$$= \frac{2c}{n}\left[\log x\left(\frac{x^2}{2}\right) - \int \frac{1}{x}\cdot\frac{x^2}{2}\,dx\right] + \theta(n)$$

$$= \frac{2c}{n}\left[\frac{x^2}{2}\log x - \left(\frac{x^2}{4}\right)\right]_{x=2}^{x=n} + \theta(n)$$

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$$= \frac{2c}{n}\left[\left(\frac{n^2}{2}\log n - \frac{n^2}{4}\right)' - \underbrace{\left(2\log_e 2 - 1\right)}_{>0}\right] + \Theta(n)$$

$\because$ subtracted

$$\leq \frac{2c}{n}\left[\frac{n^2}{2}\log n - \frac{n^2}{4}\right] + \Theta(n)$$

$$= cn\log n + \underbrace{\left(\Theta(n) - \frac{cn}{2}\right)}$$

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$\leq cn \log n$

$\in O(n \log n)$

↳ Choose c large enough. such that $\Theta(n) - \frac{cn}{2}$ is negative.

(ie $\frac{cn}{2}$ dominates $\Theta(n)$)

$\Rightarrow E(T(n)) \in O(n \log n)$

(Average Case)

Result: Any decision tree sorting n element has height $\Omega(n \log n)$

OR

Any Comparison sort algorithm requires $\Omega(n \log n)$ Comparisons in the worst Case

Proof:

$\{a_1, a_2, a_3\}$ — 3! "Permutation"

↗ Arrange (sorting)

6 possibilities

$a_1\ a_2\ a_3$

$a_1\ a_3\ a_2$

$a_2\ a_1\ a_3$

$a_2\ a_3\ a_1$

$a_3\ a_1\ a_2$

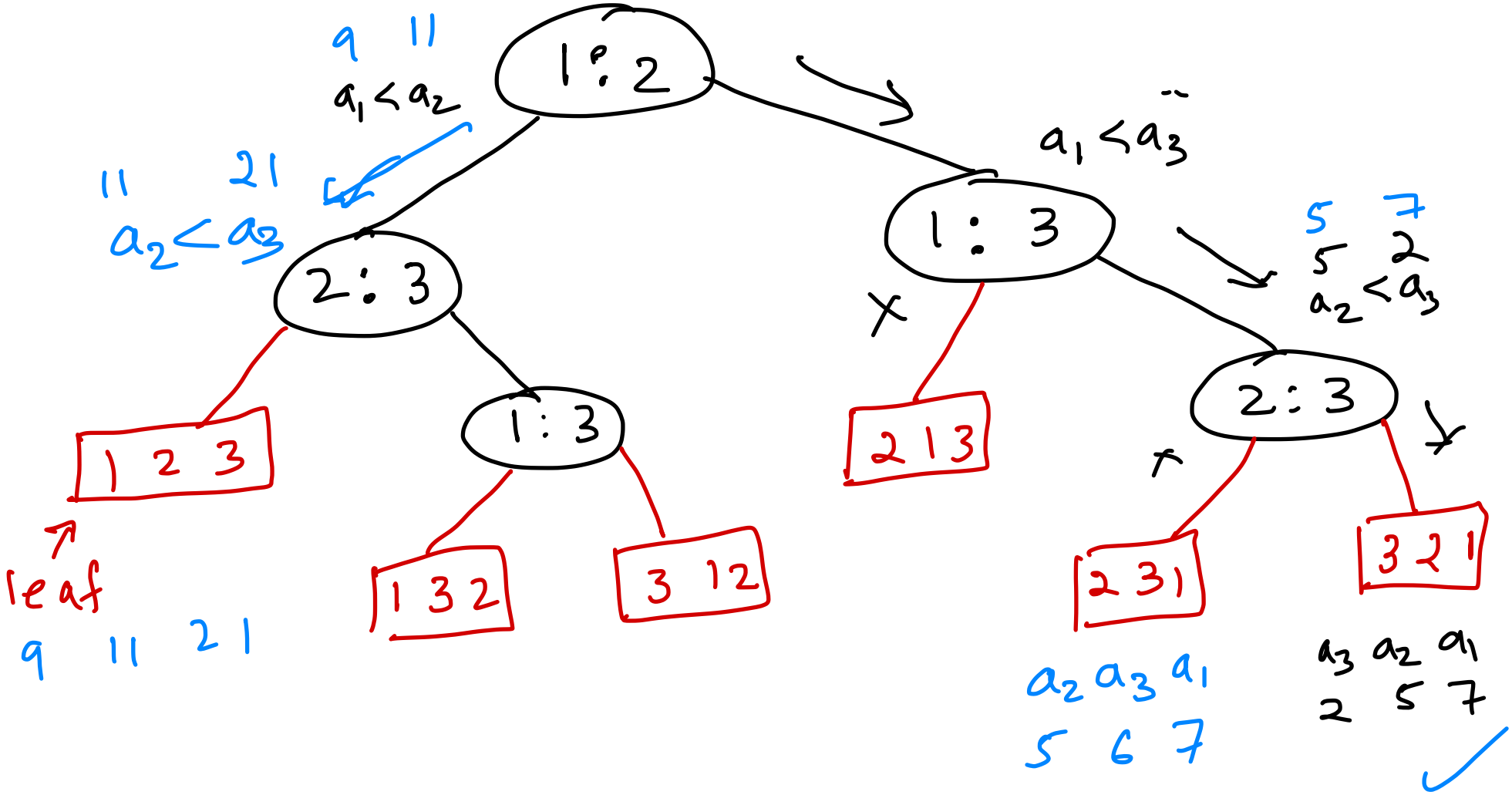
$a_3\ a_2\ a_1$

→ Leaves in a Decision Tree and one of the leaves gives the sorted list.

9 11 21

Input:

$< a_1, a_2, a_3 >$



- Any comparison algorithm is basically follows the framework given by the decision tree model:

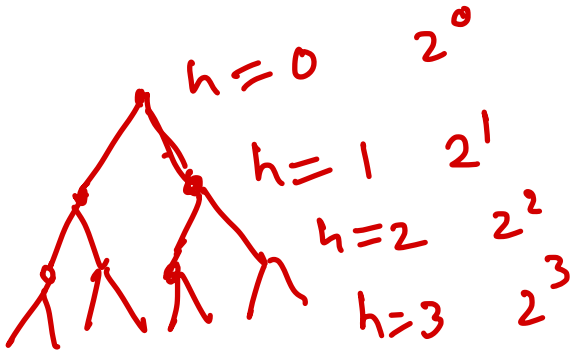
- The worst-case running time
  = longest path in the decision tree model
  = height of the decision tree
  = h.
- The height of the decision tree must be such that all leaves are accomodated (covered).
- Total number of leaves = $n!$

No. of leaves at height at most h $\geq n!$

$\uparrow$?

$2^h \geq n!$

$h = ?$



by inserting lowest possible integer on a Computer if all space at height h is not filled.

$$h \geq \log n!$$

assume $n \to$ even otherwise add 1.

$$n! = 1 \cdot 2 \cdot 3 \cdots n$$

$$\geq \left(\frac{n}{2}\right)\left(\frac{n}{2}+1\right) \cdots n.$$

$$\geq \underbrace{\left(\frac{n}{2}\right)\left(\frac{n}{2}\right) \cdots \left(\frac{n}{2}\right)}_{(n/2) \text{ times.}}$$

$$= \left(\frac{n}{2}\right)^{n/2}$$

$$n! \geq \left(\frac{n}{2}\right)^{n/2}$$

$$\log n! \geq \frac{n}{2} \log \frac{n}{2}$$

$$= \frac{n}{2}\left[\log n - \log 2\right] \in \Omega(n \log n)$$

$$\log n! \geq \frac{n}{2}\log n - \frac{n}{2}\log 2$$

$$= \left(\frac{n}{4}\log n + \frac{n}{4}\log n\right) - \frac{n}{2}\log 2$$

$$= \frac{n}{4}\log n + \left(\frac{n}{4}\log n - \frac{n}{2}\log 2\right)$$

Comparison:

$$\lim_{n\to\infty} \frac{\frac{n}{2}\log n - \frac{n}{2}\log 2}{n\log n} = \frac{1}{2} - \frac{1}{2}\lim_{n\to\infty}\frac{\log n}{n\log n}$$

$\to 0$

$$= \frac{1}{2} \neq 0.$$

$\log n! \in \Omega(n \log n)$

$h \geq \log n! \in \Omega(n \log n)$.

→ Proof:

This proves that decision tree sorting n element has height $\Omega(n \log n)$

OR

Any Comparison sort algorithm requires $\Omega(n \log n)$ Comparisons in the worst-case.

Since Quicksort, Mergesort, Insertion sort, Bubblesort etc. use comparisions their minimum running time is bounded below by $n \log n$.

This also proves that Mergesort, randomized Quicksort are asymptotically optimal. (i.e. we found upper bound as well as lower bound as $n \log n$)

($\in \Theta(n \log n)$).

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